

Generative design of steel-prestressed concrete hybrid wind turbine tower based on machine learning and multi-objective optimization

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ABSTRACT

The steel-prestressed concrete hybrid tower is an innovative high-performance structure with significant potential to enhance wind energy efficiency. However, the design and optimization of such towers remain complex, often requiring time-consuming iterative finite element (FE) simulations and lacking intelligent methods to support the design process. This paper presents a generative design framework for hybrid towers based on machine learning models and multi-objective optimization algorithms. Initially, a FE analysis model is employed to generate structural responses of the hybrid towers under specified parameters, creating a dataset for machine learning model development. Then, classification and regression methods from machine learning models are utilized to learn the mapping between structural parameters and responses, thereby replacing traditional FE simulations to quickly evaluate each design scheme. Subsequently, the developed machine learning model is integrated into the NSGA-II optimization algorithm, using construction costs and annual energy production as dual objective functions to obtain optimal design solutions. A case study demonstrates that the generative design framework can explore and evaluate design alternatives effectively, achieving an optimal balance between construction costs and energy production for hybrid towers.

1. Introduction

As global emphasis on sustainable development increases, wind energy has gained widespread recognition as a clean renewable energy source. It effectively reduces greenhouse gas emissions while providing stable energy supplies for various countries. In recent years, wind turbines have played a crucial role in enhancing wind energy utilization efficiency, establishing wind power as a significant component of global renewable energy development [1–3].

Currently, onshore wind turbines dominate the market, especially in regions abundant in wind resources, where substantial development potential exists. The trend towards larger wind turbines is becoming increasingly apparent. However, traditional steel wind turbine towers struggle to meet the rising demands of hub heights and loads, often leading to buckling and resonance issues. To ensure the strength and stiffness of towers, wall thickness and diameter must be increased, complicating manufacturing and resulting in transportation height restrictions [4,5], which has driven researchers to explore new tower design solutions. Given these challenges, steel-prestressed concrete hybrid tower (SPCHT) has gained attention for their unique structural

advantages (Fig. 1). Comprising lower concrete tower segments, transition segments, and upper steel tower segments, SPCHT significantly enhances load-bearing capacity while effectively reducing weight, lowering material consumption, and improving construction efficiency. These benefits align well with the requirements of large-scale wind turbine projects [6].

As the supporting structure for high-capacity wind turbines, the construction cost of tower structures accounts for approximately 20 % of the total expense of wind turbines [7,8]. Therefore, optimizing tower structure design is increasingly crucial for achieving cost-competitive electricity generation. Unfortunately, current traditional optimization methods are time-consuming, labor-intensive, and often inefficient, making it challenging to meet the rapidly evolving market demands. In contrast, utilizing intelligent optimization design methods that combine computer technology with advanced algorithms enables the rapid development of structural optimization solutions tailored to project objectives.

Researchers have extensively studied the optimization design methods of wind turbine towers using heuristic algorithms. Chen et al. [9] considered constraints such as stress, frequency, and slenderness

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ratio, employing particle swarm optimization to optimize the lattice tower shape and cross-sectional dimensions, which resulted in a significant reduction in steel consumption. Gentils et al. [10] optimized the cross-sectional dimensions of offshore wind turbine towers through finite element (FE) analysis and genetic algorithms, resulting in a significant reduction in structural mass while emphasizing natural frequency and fatigue as critical design factors. Lana et al. [11] conducted a parametric FE analysis model for prestressed concrete wind turbine towers, coupling it with genetic algorithms to create an innovative structural integration optimization model. Chen et al. [12] obtained optimized design parameters that meet performance requirements using an improved genetic algorithm for SPCHTs. Huang et al. [13] derived an analytical formula for the natural frequency of SPCHT and utilized the Matlab genetic algorithm toolbox to achieve a 12.7 % reduction in construction costs of a 160-meter hybrid tower. Li et al. [14] proposed a new parallel PSO algorithm to obtain the geometric dimensions of SPCHTs, using the levelized cost of energy (LCOE) as the objective function, significantly improving computational efficiency. Additionally, in our previous research [15], the author integrated the OpenSees program with an improved evolutionary algorithm, comprehensively addressing performance and boundary constraints for hybrid towers and optimizing geometric dimensions, thereby overcoming issues related to low efficiency of solid FE models and inaccuracies in analytical formulas. While numerous studies have focused on the optimization design of hybrid towers, there remain areas for improvement in both optimization objectives and algorithms.

In terms of optimization objectives, current research predominantly focuses on reducing the construction costs of tower structures, often overlooking the impact on annual energy production (AEP). Specifically, scholars tend to concentrate on optimizing parameters such as wall thickness and diameter while neglecting the optimization of hub height.

Hub height is a critical parameter for wind turbines. The greater the hub height, the more wind energy the turbine can capture, generating higher energy production. However, increasing hub height also leads to higher manufacturing costs, more stringent construction requirements, and more significant transportation challenges. Therefore, to accommodate the varying wind resource conditions across different regions, selecting an appropriate hub height necessitates a comprehensive evaluation of both energy generation and investment costs, ensuring that investment optimization is achieved.

In terms of optimization algorithms, while heuristic algorithms possess strong global search capabilities, they require repeated calls to FE simulation models during the optimization iteration process to evaluate various design options, resulting in lengthy optimization times. Therefore, more efficient optimization methods need to be developed in the preliminary design phase of tower structures. Recently, machine learning methods have emerged as effective data-driven strategies, attracting considerable attention from researchers. These methods have seen rapid development in damage identification, pattern recognition, and parameter prediction within civil engineering [16–20]. Integrating trained machine learning models into optimization algorithms can serve as surrogate models for the time-consuming FE simulation models during optimization iterations, allowing for more efficient exploration of design solutions that meet project goals and significantly enhancing design efficiency. Fig. 2 shows surrogate-model-based and FE-model-based optimization framework. This research approach has already been applied to optimization problems in truss structures [21], steel frame cross-section optimization [22], and telescope shell structure optimization [23].

Currently, some researchers have applied machine learning to the optimize of wind turbine towers. For instance, Dai et al. [24] employed a BPNN and the NSGA-II algorithm to design the steel wind turbine tower,

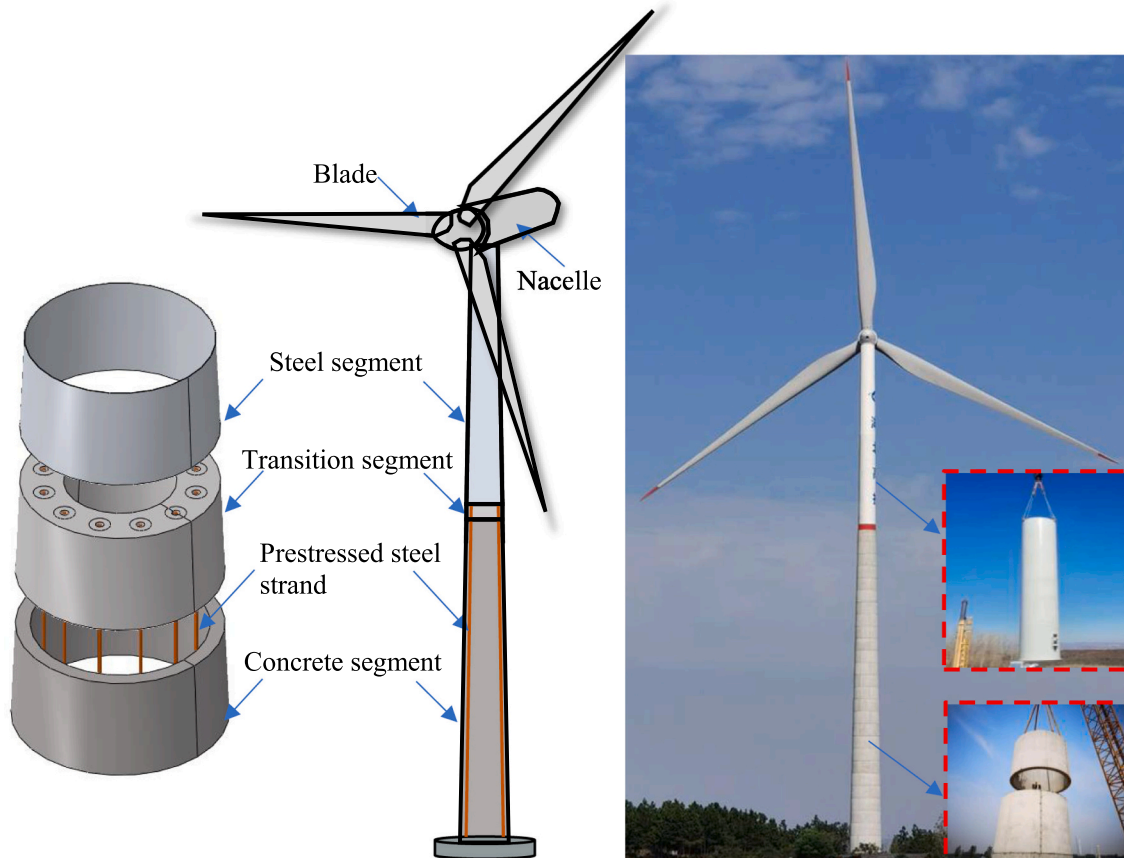


Fig. 1. Steel-concrete hybrid wind turbine tower.

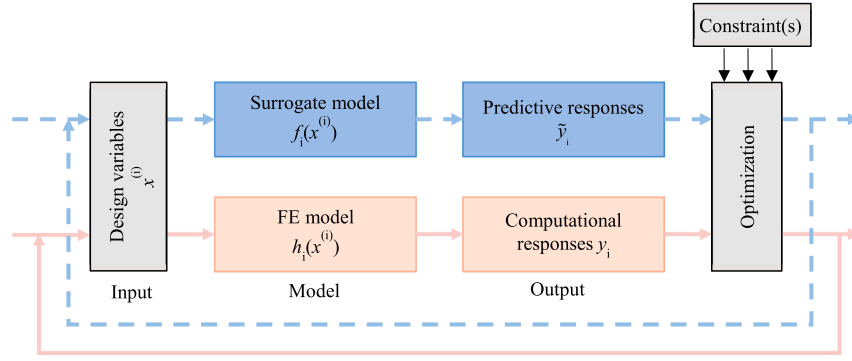


Fig. 2. Surrogate-model-based and FE-model-based optimization framework.

using mass and top displacement as objective functions. Anda et al. [25] utilized artificial neural networks to predict the reliability indicators of onshore wind turbine steel towers and obtained optimal design solutions based on a multi-objective particle swarm algorithm. However, research on more complex hybrid towers remains limited.

To address the aforementioned challenges, this paper integrates machine learning models with multi-objective optimization algorithms to propose a novel generative design framework for SPCHTs. The hub height is also included as an optimization variable, allowing for a comprehensive consideration of minimizing construction costs and maximizing AEP in the optimization of the tower's geometric parameters. In terms of optimization algorithms, a high-precision machine learning model is developed to predict the structural responses, enabling

the rapid evaluation of each design solution generated by the optimization algorithm. This approach effectively mitigates the time-consuming issue associated with repeatedly calling the FE model during the optimization process.

2. Generative design framework

In this study, a generative design framework for SPCHTs is presented based on machine learning and multi-objective optimization algorithms. The framework aims to balance minimum construction costs and maximum AEP by rapidly and accurately exploring and assessing automatically generated design options. Fig. 3 illustrates an overview of the framework, which includes three main sub-models: database

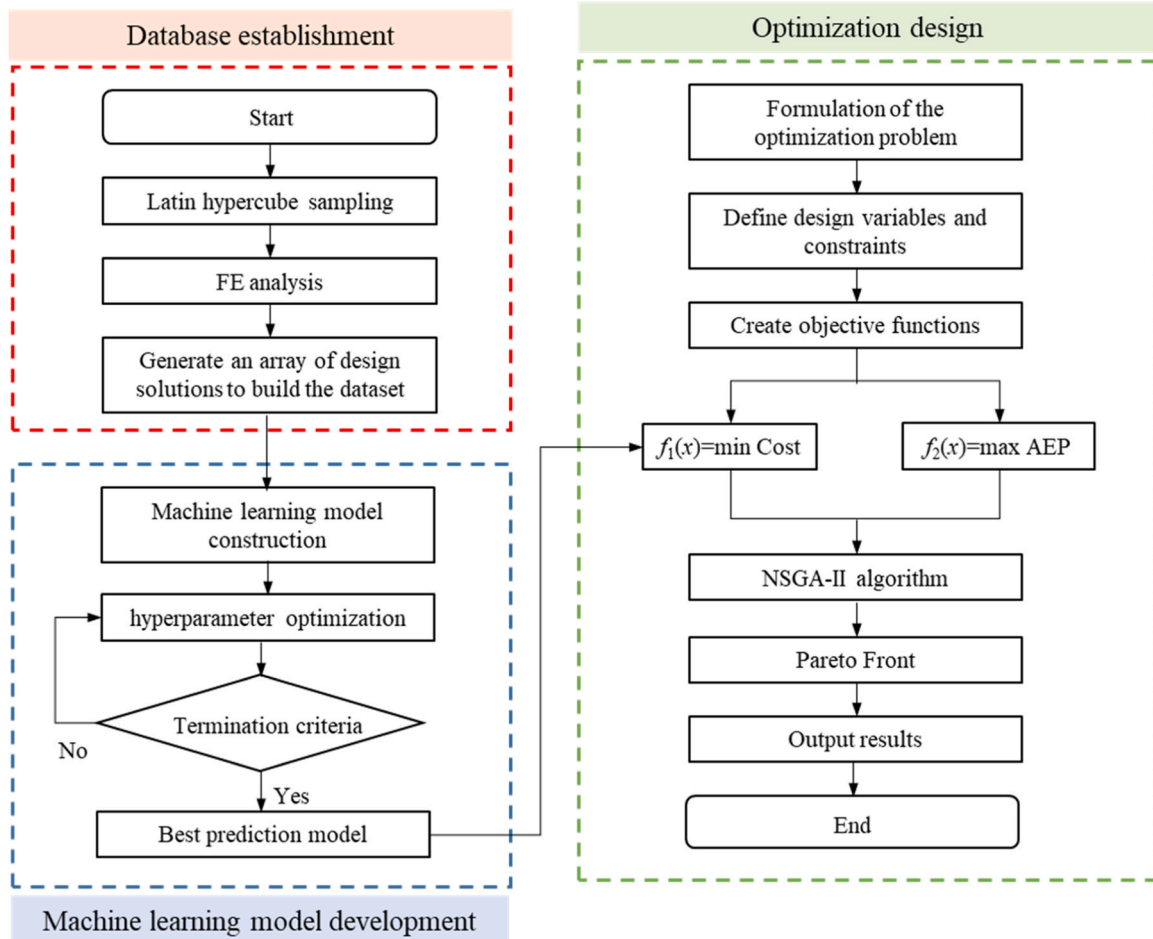


Fig. 3. The framework of the hybrid towers generative design.

establishment, machine learning model development, and multi-objective optimization design. The database establishment phase creates a dataset for machine learning model development based on FE models. In the machine learning model development phase, classical machine learning models are developed to learn the mapping between structural parameters and responses, thereby replacing traditional FE simulations for the rapid evaluation of each design option. The multi-objective optimization algorithm provides optimal design solutions under given constraints and objectives. The entire framework operates on the open-source computational platform Python.

3. Database establishment

3.1. Wind load calculation

To accurately calculate the wind loads acting on wind turbines, it is essential to simulate wind speed. Generally, wind speed consists of two components: the mean wind speed and the fluctuating wind speed. The mean wind speed with height is defined as [26]:

$$v(z) = v_{\text{hub}} \left(\frac{z}{h_{\text{hub}}} \right)^{\alpha} \quad (1)$$

Where $v(z)$ represents the mean wind speed at height z ; v_{hub} and h_{hub} denote the mean wind speed and height at the hub, respectively; and α is the wind shear coefficient.

The Kaimal spectrum is adopted to analysis the turbulent components of wind speed in the wind field [27]. When simulating the time series of fluctuating wind speed at various points using the harmonic superposition method, the along-wind fluctuating wind speed time series is expressed as:

$$v_j(t) = \sqrt{2(\Delta\omega)} \sum_{m=1}^j \sum_{l=1}^N |H_{jm}(\omega_{ml})| \cos[\omega_{ml}t - \theta_{jm}\omega_{ml} + \phi_{ml}] \quad (2)$$

Where N represents the number of divisions of the fluctuating wind frequency; j denotes the number of points used for calculations. $\Delta\omega = \omega/N$ is the frequency increment; ω_{ml} is defined as $(l-1) \cdot \Delta\omega + (m/j) \cdot \Delta\omega$; ϕ_{ml} is independently and uniformly distributed within the interval $[0, 2\pi]$; $H_{n \times n}(\omega)$ is the Cholesky decomposition of the power spectral matrix $S_{n \times n}(\omega)$; $\theta_{n \times n}(\omega)$ represents the phase angle of $H_{n \times n}(\omega)$. Additionally, the element in the i th row and j th column of the power spectral matrix $S_{n \times n}(\omega)$ is given by:

$$S_{ij}(\omega) = \sqrt{S_{ii}(\omega)S_{jj}(\omega)} S_{ij} \text{Coh}_{ij} \quad (3)$$

Where Coh_{ij} represents the coherence coefficient between two spatial points i and j .

Based on Eqs. (1–3), a program is developed to simulate the mean and fluctuating wind speeds at various spatial points, thereby generating the required wind speed time series.

The wind load on the tower primarily consists of the loads from the wind turbine and the tower itself. IEC 61400-1 [28] standards specify load calculations for wind turbines under various operating conditions, including normal power generation, shut down, start up, and fault scenarios. Strictly speaking, a comprehensive load simulation should be performed for each optimization iteration, ensuring the thoroughness and accuracy of the structural design. Considering all conditions in every iteration would significantly increase computational cost and reduce optimization efficiency. Therefore, during the optimization phase, the most representative operating conditions (DLC 1.1 and DLC 6.1) are considered to obtain a reliable optimized design within a reasonable computational resource limit. In this study, the aerodynamic thrust force applied at the tower top adopts a thrust coefficient method for wind turbine load calculation to improve computational efficiency. It is worth noting that the thrust coefficient is not modeled as a constant value, but

rather derived from aeroelastic simulations performed conducted by the wind turbine manufacturer using the industry-standard software Bladed. The thrust force corresponding to different wind speeds is obtained from a full turbine model that includes rotor–airflow interaction and aeroelastic effects. As such, the influence of aerodynamic damping is inherently embedded in the load data. Although aerodynamic damping is not explicitly defined in the structural model, the use of the load conditions provided by the manufacturer presents a realistic approximation of the dynamic behavior of the turbine, including the damping effects associated with rotor–air coupling. For the normal operating condition of DLC 1.1, the horizontal thrust $F_{x1.1}$ acting on the rotor [29] is defined as:

$$F_{x1.1} = \frac{1}{2} C_T \rho A v^2 \quad (4)$$

Where C_T is the thrust coefficient; ρ is the air density; and A is the swept area of the blade. The thrust coefficient of the wind turbine at different wind speeds in this study is shown in Fig. 4.

The wind load on the tower is distributed along its height. According to the Chinese code [30], the wind load $q(z)$ acting on the tower is defined as:

$$q(z) = \beta_z \mu_s(z) \mu_z(z) w_0 D(z) \quad (5)$$

Where β_z is the wind vibration coefficient; $\mu_s(z)$ is the shape coefficient for the wind load; $\mu_z(z)$ is the wind pressure height variation coefficient; w_0 is the basic wind pressure; and $D(z)$ is the tower diameter.

Under the DLC 6.1 condition, where the blades are stationary or idling, the aerodynamic load $F_{x6.1}$ acting on the rotor can be expressed as:

$$F_{x6.1} = \frac{1}{2} C_d \rho v_{e50}^2 A_{\text{proj}} B \quad (6)$$

Where v_{e50} represents the extreme wind speed with a return period of 50 years; C_d is the drag coefficient of the blade [29], and A_{proj} is the blade area projected perpendicular to the wind. The wind load on the tower can similarly be calculated using Eq. (6), with the necessary adjustments to the basic wind pressure value.

For any customized hybrid tower at a specified hub height, the extreme load on the wind turbine is taken as the maximum value between normal operating and extreme conditions. Variations in long-term average wind speed and short-term fluctuating wind speed are the primary causes of fatigue loads on the turbine. For long-term average wind speed variations, the operational wind speed range for the wind turbine extends from the cut-in to the cut-out speeds, and the wind turbine will stop operating outside this range. The operational wind speed is divided into several intervals, with the midpoint of each interval used to represent the corresponding wind speed. For short-term fluctuating

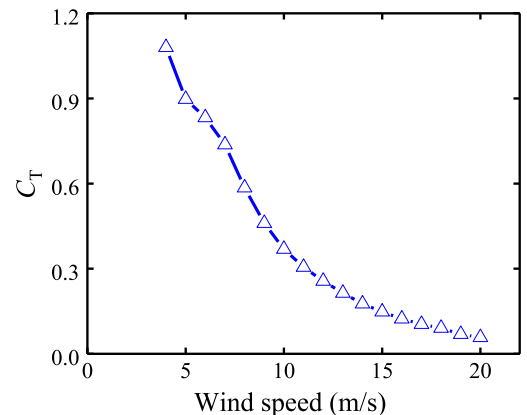


Fig. 4. Thrust coefficient at different wind speeds.

wind speed, each wind speed is assumed to last for ten minutes, and the Kaimal spectrum is employed to describe the instantaneous wind speed sequence that corresponds to the ten-minute average wind speed for each condition. For short-term fluctuating wind speed, each wind speed's duration is set at 10 min, and the Kaimal spectrum is used to describe the instantaneous wind speed sequence corresponding to the ten-minute average wind speed for each condition. This approach allows for the calculation of the extreme and fatigue loads on the turbine at any height.

In this study, the tower height is also considered an optimization variable, meaning that any changes in tower height will result in corresponding changes in the wind turbine loads. A wind turbine load database has been established to reduce the computational time required for calculating wind turbine loads during optimization. The suitable height range for SPCHTs is typically between 100 m and 180 m. Before optimization, extreme and fatigue loads for different heights were calculated in increments of 5 m and stored for easy retrieval during the optimization process.

3.2. FE model establishment

A FE model of the SPCHT has been developed using OpenSeesPy, as shown in Fig. 5. Both the concrete and steel tower segments are modeled using displacement-based beam-column elements, with the elements discretized according to the height of the steel and concrete segments. The tower element sections utilize fiber cross-sections, applying the Lobatto integration method with five integration points along the length. For concrete tower segment elements, the cross-section is discretized into 16 radial fibers and 2 circumferential fibers, whereas for steel tower segment elements, it is divided into 16 radial fibers and a

single circumferential fiber. Currently, the design of wind turbine structures is based on the principle of elastic design, which requires that the materials do not enter the plastic stage. Therefore, the material constitutive relationship is defined using an elastic constitutive model. This approach not only complies with design specifications but also effectively avoids convergence difficulties and excessive computational time caused by some individuals entering the elastoplastic stage. The unbonded prestressing steel strands are modeled with Truss elements, and initial prestress is applied using Steel02 material. The bottom nodes of the prestressing steel strands are fixed, while the top nodes are linked to the bottom nodes of the transition segment via rigid beam elements to ensure cooperative loading with the tower. To simulate the joint behavior between the concrete prefabricated segments, two nodes with the same coordinates are created at the joint and assigned a Zero Length Section element using Elastic-No Tension material that is only compressive. The Equal DOF command is applied to constrain the two nodes at the joint from experiencing relative horizontal displacement. Additionally, nodes with mass and inertia are placed at the center of the nacelle and blades, linked to the top tower node via rigid beam elements. Due to the complex structural configuration of the transition segment in practical engineering, a refined finite element analysis is required. To reduce the number of optimization variables and improve the efficiency of the preliminary tower design, the transition segment's detailed structure is not explicitly considered during modeling. Instead, it is modeled as a standard concrete tower segment. Similarly, structural verification of the transition segment is not conducted in this study.

3.3. Structural response extraction

The dynamic characteristics of the SPCHT are analyzed to determine

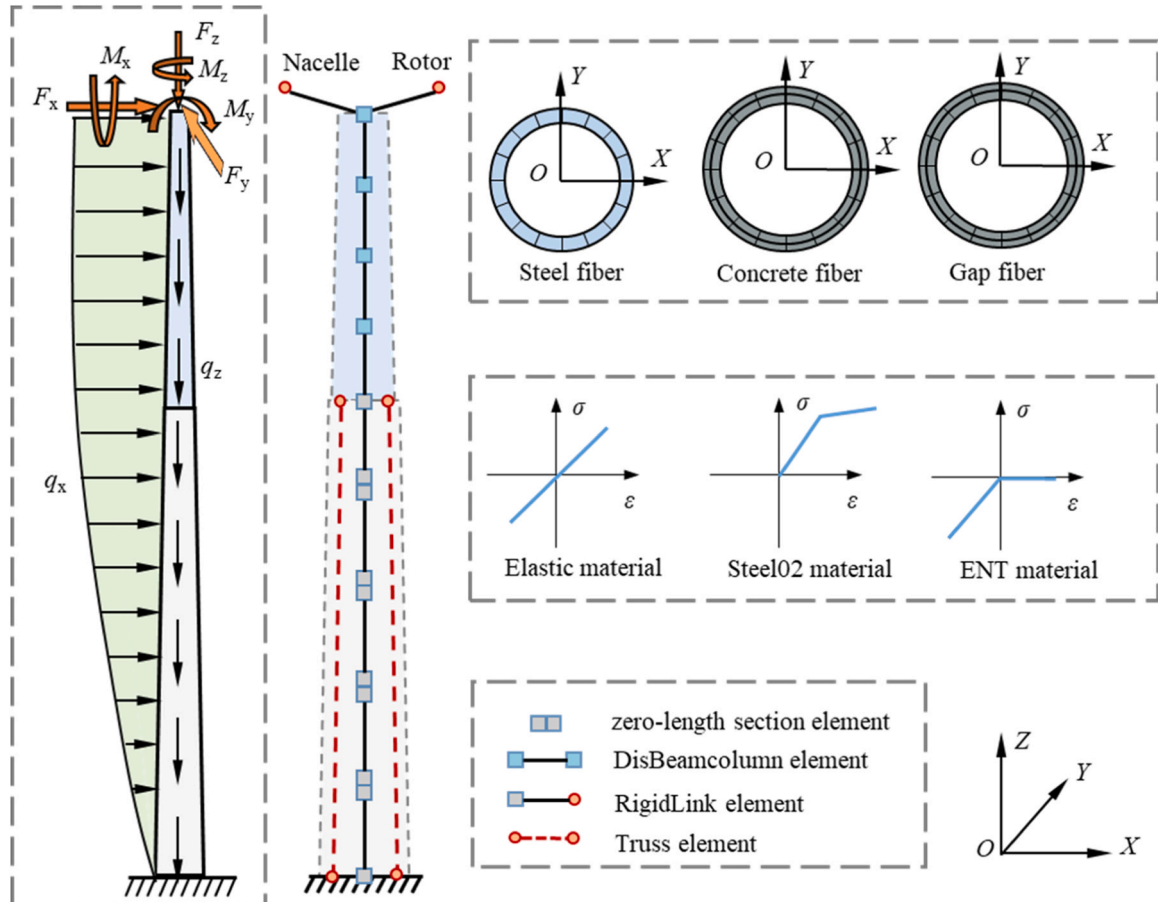


Fig. 5. OpenSeespy model of SPCHT.

the first-order frequency (f_{1st}) of the structure. Subsequently, loads corresponding to three different conditions: normal operation, ultimate limit state, and fatigue, are applied to evaluate the structural responses, respectively. Under normal operation condition, the maximum tensile stress of concrete (σ_{scr}), as well as the maximum displacement and rotation at the tower top (Δ_{max} , θ_{max}), can be directly extracted from the model. Under the ultimate limit state, the maximum stress of steel (σ_{us}), along with the maximum compressive and tensile stresses of concrete (σ_{ucc} , σ_{uct}), can also be obtained from the model. The local stability stress and overall stability check value of the steel tower segment (σ_{scr} , σ_{sos}) are calculated according to the corresponding codes [31,32]. For the fatigue condition, the maximum fatigue damages for concrete and steel (D_c , D_s) cannot be directly obtained from the FE model. Fig. 6 illustrates a fatigue damage calculation framework based on the probability distribution of wind speed, comprising the following process: (1) Stress time history curves corresponding to each wind speed are obtained through FEA; (2) Rain flow counting method is used to convert the stress time-histories into multiple stress cycles for each wind speed; (3) The total fatigue damage D_n for the tower at each wind speed level during its lifespan is calculated according to linear cumulative damage theory; (4) The probability of each wind speed is treated as a damage weighting factor ϕ_n for the structure at that wind speed. The fatigue damage values D_n obtained in step (3) are linearly accumulated to yield the total damage value D , as expressed in Eq. (7).

$$D = \phi_1 D_1 + \phi_2 D_2 + \dots + \phi_n D_n = \sum_{i=1}^n \phi_n D_n \quad (7)$$

3.4. Dataset establishment

Based on the FE model described above, a dataset for machine learning model development is created. Latin Hypercube Sampling (LHS) is used to sample the design variables of the SPCHT. The design variables include the bottom and top diameters of the concrete segment (D_{cb} , D_{ct}), the bottom and top wall thicknesses of the concrete segment (t_{cb} , t_{ct}), the bottom diameter of the steel segment D_{sb} , the bottom and top wall thicknesses of the steel segment (t_{sb} , t_{st}), the area of the pre-stressing steel strands (A_s), the height of the concrete segment (H_c), and

the total height of the tower (H). The top diameter of the steel tower is not treated as an optimization variable because it is constrained by the nacelle-tower interface requirements. For each set of design variables (model inputs), a FE analysis is conducted to obtain necessary structural responses (model outputs). Each set of parameters represents a design configuration. Each set of design variables and structural responses is compiled into a one-dimensional vector. A database matrix required for the machine learning model is formed by repeating this process N times, as shown in Fig. 7.

4. Machine learning model development

4.1. Machine learning models

Machine learning models can adaptively grasp the mapping relationships between inputs and outputs from data, offering higher identification accuracy than traditional empirical formula fitting. This study develops machine learning models for the relationship between structural parameters and structural responses using three classic algorithms: Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Extreme Gradient Boosting Trees (XGBoost).

SVM maps linearly inseparable samples from a low-dimensional input space to a high-dimensional feature space using non-linear mapping algorithms [33]. It then seeks an optimal hyperplane in this space to maximize the margin between classes. For classification tasks, its decision function is defined as follows:

$$f(x) = \text{sgn} \left(\sum_{i=1}^l \alpha_i y_i \kappa(x_i, x) + b \right) \quad (8)$$

Where l represents the number of support vectors; α_i denotes the Lagrange coefficient for the i -th support vector; y_i is the class label for the i -th sample; κ is the kernel function; x is the feature parameter vector; x_i corresponds to the feature parameter vector of the i -th support vector; and b is the bias term of the decision boundary.

ANN consist of an input layer, hidden layers, and an output layer, with each layer comprising neurons that perform different functions [34]. The neurons in the input layer receive information (feature

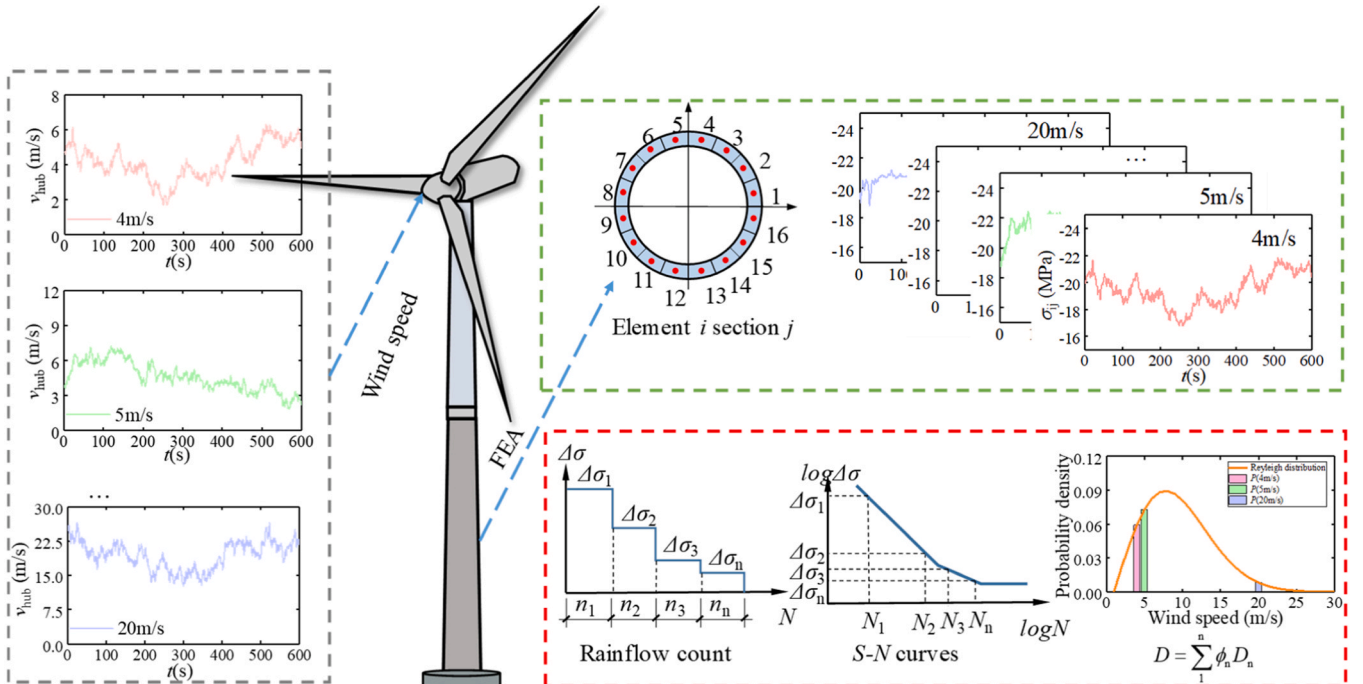


Fig. 6. Fatigue damage calculation framework of SPCHT.

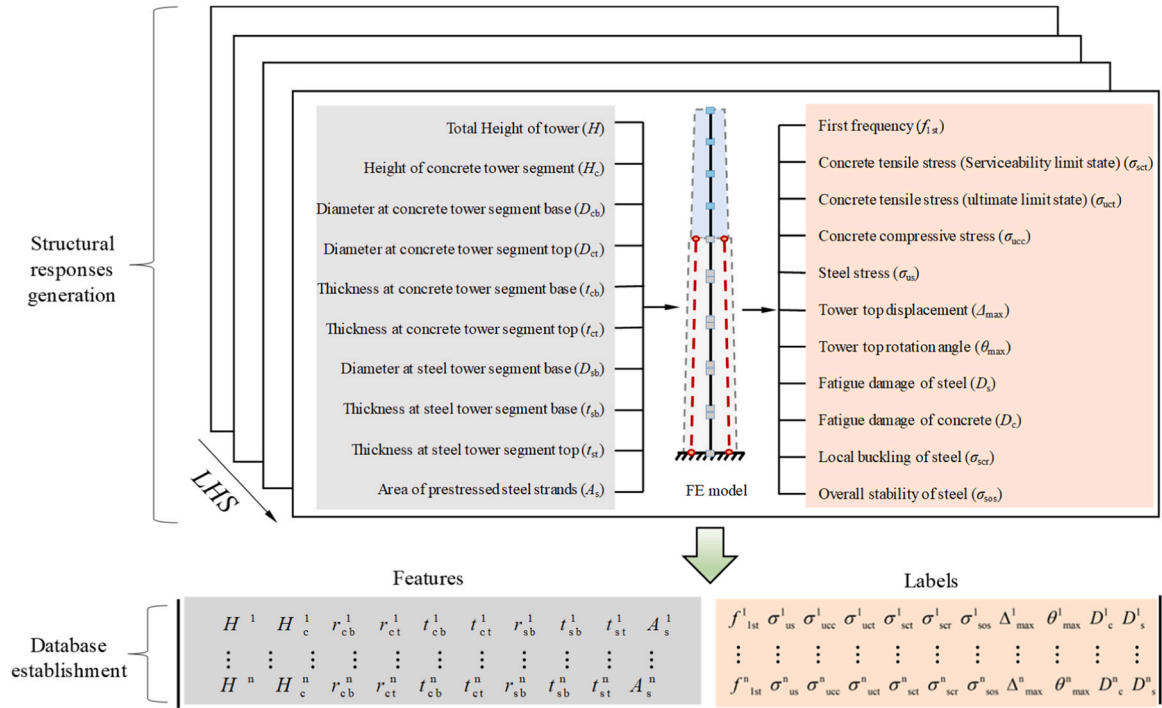


Fig. 7. Database establishment of SPCHTs.

values), which are then passed to the hidden layer through weights and activation functions. The outputs of each neuron in the hidden layer are computed based on these weights and thresholds, ultimately producing the final output in the output layer. The output of the i th neuron in the l th layer is given by:

$$o_i^{(l)} = f\left(\sum_{j=1}^{N^{(l-1)}} w_{ij}^{(l-1)} o_j^{(l-1)} + b_i^{(l-1)}\right) \quad (9)$$

Where x_i represents the i th feature of the sample's feature parameter vector x ; w_{ij} denotes the connection weight between the j th node of the $(l-1)$ th layer and the i th node of the l th layer; b_{ij} is the bias of the i th node in the $(l-1)$ th layer; $N^{(l-1)}$ indicates the number of nodes in the $(l-1)$ th layer; and f represents the activation function.

XGBoost is a leading ensemble learning algorithm known for its high accuracy and generalization capability. The fundamental concept of XGBoost is sequential learning, where each new decision tree fits the residuals of the previously integrated decision trees, thereby reducing

the overall model residuals [35]. XGBoost employs decision trees as base learners, and it integrates the predictions from multiple decision trees to generate the final output, as expressed in the following equation:

$$o_i = \sum_{k=1}^K f_k(x_i) \quad (10)$$

Where o_i is the predicted value for the i th target; K is the trees number; and f_k is the function representing the structure and leaf weights of the k -th tree.

Fig. 8 presents three different graphical representations of various machine learning models. Machine learning models can be categorized into regression algorithms and classification algorithms based on the type of problem being addressed. Regression algorithms can be employed to model the responses for parameters such as tower frequency, stress, and deformation metrics. In contrast, due to the significant discretization of design variables associated with fatigue damage indicators, classification algorithms are used to determine whether fatigue failure has occurred. During the optimization iteration process, the

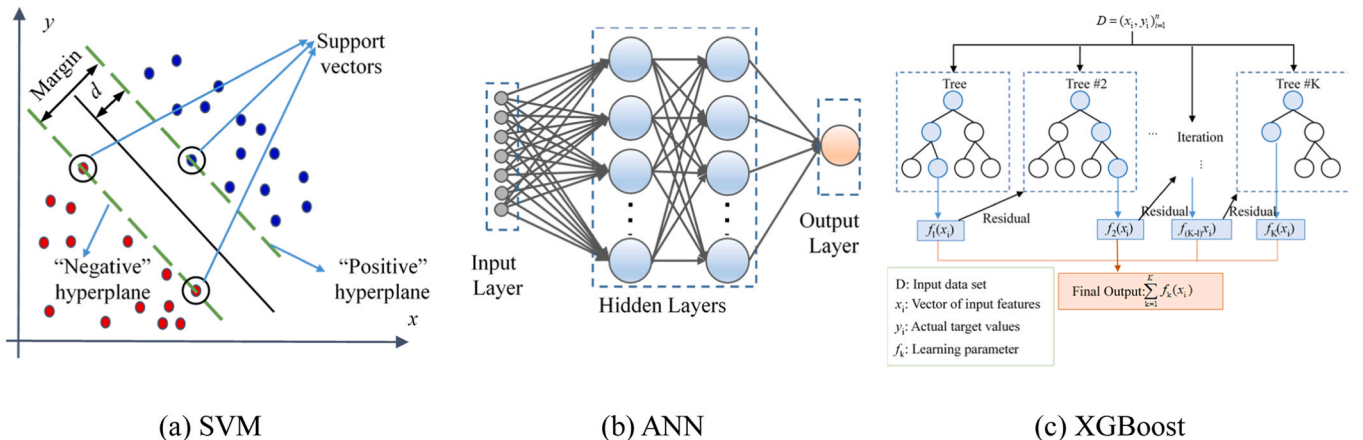


Fig. 8. Graphical representations of various machine learning models.

focus is often on whether the design has experienced fatigue failure, rather than on the specific values of fatigue damage. Therefore, labeling the output features of samples based on whether the cumulative fatigue damage exceeds 1 is a practical approach that can significantly enhance the accuracy of design assessments.

4.2. Model training process

Based on the workflow shown in Fig. 3, a hybrid tower structure response prediction model is established using machine learning models. The training of the machine learning model in this study was conducted using the scikit-learn platform and performed on a personal computer equipped with an Intel(R) Core(TM) i7-12700F CPU @ 2.10 GHz. The process is detailed as follows: (1) Dataset Splitting: The dataset is divided into training and testing sets in a 7:3 ratio. (2) Model Training: Each model is trained using the training set, applying ten-fold cross-validation and grid search strategies for hyper-parameter optimization. The training aims to minimize the mean absolute error (MAE), ultimately yielding a prediction model with the globally optimal hyper-parameter combination. (3) Performance Evaluation: The model obtained in step (2) is validated using the testing set. The prediction accuracy of the regression algorithm is quantitatively assessed using metrics such as the coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE), as expressed in Eqs. (11–13):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - p_i)^2}{\sum_{i=1}^n (y_i - y_m)^2} \quad (11)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - p_i)^2} \quad (12)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - p_i| \quad (13)$$

Where y_m represents the sample mean; y_i and p_i denote the measured and predicted values for the i th target, respectively; and n is the total number of samples.

The prediction accuracy of the classification algorithm is quantitatively assessed using metrics such as precision, recall, and classification accuracy, as expressed in Eqs. (14–16):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (14)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (15)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (16)$$

Where TP and FP represent the true positives and false positives, respectively; FN and TN denote the false positives and true negatives, respectively.

Through the above process, a machine learning model suitable for predicting tower structure responses is developed, as shown in Fig. 9. When using the model, providing the tower structure parameters allows for the prediction of the tower structure responses.

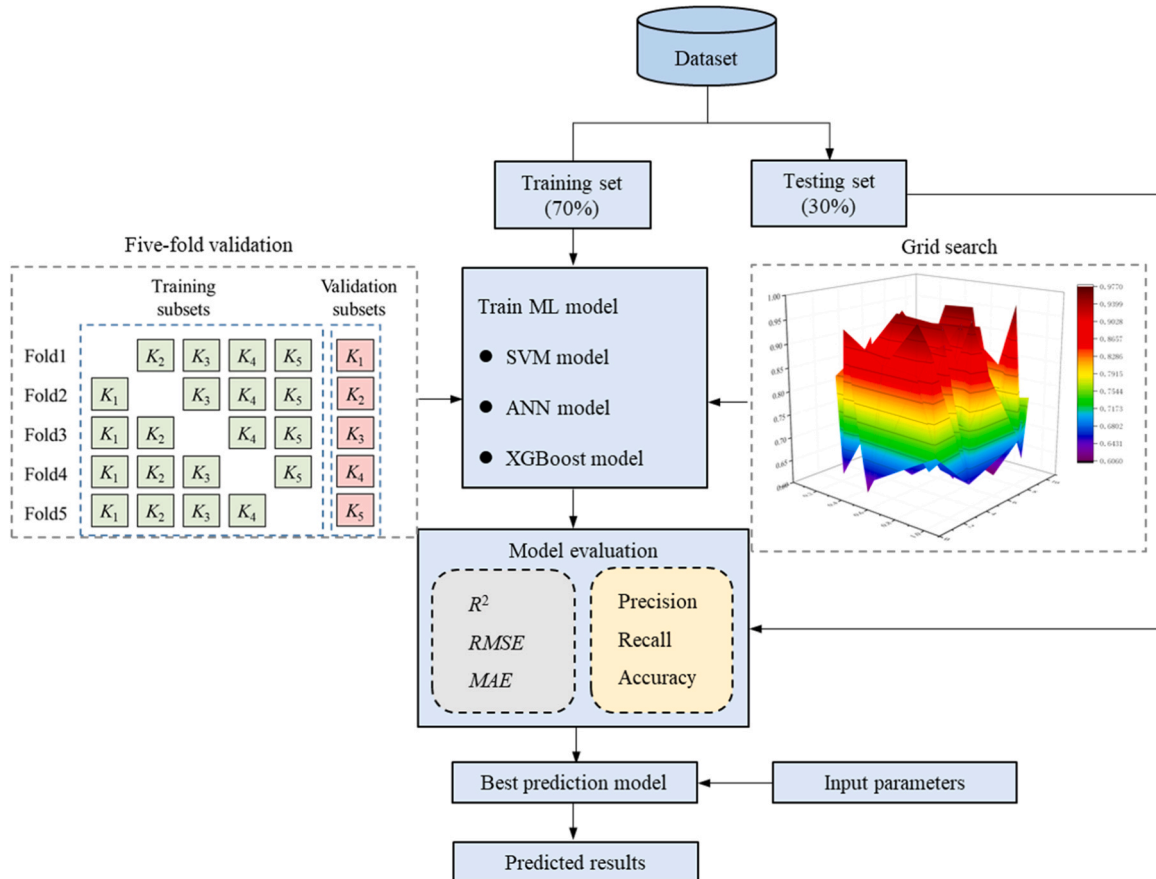


Fig. 9. Machine learning model training process.

5. Multi-objective optimization

5.1. Objective function

This study establishes two conflicting objective functions for the optimization design of hybrid towers: minimizing construction costs f_1 and maximizing AEP f_2 .

5.1.1. Minimizing construction costs f_1 based on performance constraints

Taking into account material costs and various design constraints, the construction cost of the SPCHT is defined as follows:

$$f_1 = \sum_{i=1}^m C_i V_i + \sum_{i=1}^n p_i g_i \quad (17)$$

Where V_i and C_i denote the consumption volume and unit price of corresponding material, respectively; p_i is the constraint penalty value, set to 100; g_i is the constraint value, taking a value of 0 when constraints are satisfied and g_i when they are not. A smaller f_1 value indicates better economic efficiency while satisfying all design constraints.

5.1.2. Maximizing AEP f_2 based on wind speed probability distribution

Based on the Rayleigh distribution function $f(v)$ [28] of wind speed and the power curve function $p(v)$ of wind turbines [36], a mathematical model for AEP is established using Eq. (18):

$$f_2 = \int_{v_{in}}^{v_{out}} p(v) f(v) \cdot N_h dv \quad (18)$$

Where N_h is the number of hours per year, taken as 8760 h.

The probability distribution curve function $f(v)$ of wind speed is expressed as:

$$f(v) = \frac{\pi v}{2 v_{ave}^2} \exp \left[-\pi \left(\frac{v}{2 v_{ave}} \right)^2 \right] \quad (19)$$

Where v_{ave} is the average wind speed at the reference hub height.

The output power curve function $p(v)$ of a wind turbine can be expressed as:

$$p(v) = \begin{cases} 0 & v < v_{in} \text{ or } v > v_{out} \\ \frac{1}{2} \rho A C_p v^3 & v_{in} < v < v_r \\ P_r & v_r < v < v_{out} \end{cases} \quad (20)$$

Where v_{in} , v_{out} , and v_r are cut-in wind speed, cut-out wind speed, and rated wind speed, respectively; C_p is the power coefficient; ρ is the air density; A is the swept area of the blade; P_r is the rated power.

According to the wind shear exponent model of Eq. (1), average wind speed varies with height. Based on these relationships, a connection can be established between the objective function of maximum AEP and the optimization variables of the tower.

5.2. Constraint condition

In accordance with tower design specifications, a set of multi-criteria performance evaluation constraints has been established, encompassing vibration, stress, deformation, buckling, and fatigue. Specifically, the first-order frequency of the hybrid tower must maintain a safety margin of 10 % from the excitation frequencies (f_{1p} , f_{3p}) during the operational process of the wind turbine rotor to prevent resonance. Under ultimate limit state conditions, σ_{us} , σ_{ucc} , and σ_{uct} must not exceed the respective material design strength values. In serviceability conditions, tensile stress is not permitted in the concrete section. Δ_{max} at the tower top should remain within 1 % of the total height, and θ_{max} should not exceed 5°. Local and overall instability is not permitted in the steel tower. Additionally, the cumulative fatigue damage of both steel and concrete

must be assessed to prevent fatigue failure during service. Furthermore, the optimization variables must satisfy a range of performance constraints aligned with design specifications, as well as boundary constraints related to practical processing and transportation requirements. Considering transportation limitations, the diameter of the steel segment must be less than 4500 mm. To accommodate standard cylindrical sections, the wall thickness of the steel segment must be less than 40 mm [37]. The wall thickness of the concrete segment must not be less than 180 mm [38]. Table 1 lists all performance constraints for the SPCHT design variables.

5.3. NSGA-II algorithm

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is a popular multi-objective evolutionary algorithm known for its low complexity, fast execution speed, and good convergence properties. Built on the principles of genetic algorithms, NSGA-II incorporates three additional operations: fast non-dominated sorting, crowding distance calculation, and elitism preservation [39,40]. The basic steps of the NSGA-II algorithm are: (1) Initial Population Generation: Define the maximum number of iterations Gen_{max} and the population size N . Randomly initialize the population P within the value ranges of each individual's dimensions. (2) Population Evolution: Apply selection, crossover, and mutation operations to population P to create a new population Q . Merge populations P and Q to form population R , with a total size of $2N$. (3) Fast Non-dominated Sorting: For each individual in population R , compute the number of individuals it dominates (n_p) and the set of individuals dominated by it (S_p). (4) Crowding Distance Calculation: For each non-dominated front, calculate the crowding distance for each individual within that front. (5) New Population Construction: Sort the individuals in the non-dominated sorting set N_D based on their rank. To ensure diversity and elitism, prioritize individuals from different classification subsets, and within the same subset, prioritize those with larger crowding distances, continuing until the specified population size N is achieved. (6) Termination Conditions: If the pre-determined number of iterations or other termination criteria (such as solution quality) is met, the algorithm terminates. Otherwise, it returns to step 3. (7) Output: Upon termination, the non-dominated solutions from the final generation are output as an approximation of the Pareto front. The procedure of the NSGA-II optimization algorithm is shown in Fig. 10.

Table 1
Design constraints for SPCHT.

Symbols	Descriptions	Constraints	Units
f_{1st}	First-order natural frequency	$\leq 0.9 f_{1p}$ or $1.1 f_{1p} \leq f_{1st}$	Hz
σ_{us}	Maximum steel stress	$\sigma_{us}^u \leq f$	MPa
σ_{ucc}	Maximum concrete compressive stress	$\sigma_{ucc}^u \leq f_c$	MPa
σ_{uct}	Maximum concrete tensile stress (ultimate limit state)	$\sigma_{uct}^u \leq f_t$	MPa
σ_{sct}	Maximum concrete tensile stress (serviceability limit state)	$\sigma_{sct}^s \leq 0$	MPa
σ_{scr}	Maximum local stability stress of steel	$\sigma_{scr} \leq \sigma_{cr}$	MPa
σ_{sos}	Maximum overall stability check value of steel	$\sigma_{sos} \leq 1$	-
Δ_{max}	Maximum displacement	$\Delta_{max} \leq H/100$	mm
θ_{max}	Maximum angle	$\theta_{max} \leq 5$	°
D_c	Maximum fatigue damage of concrete	$D_c \leq 1$	-
D_s	Maximum fatigue damage of steel	$D_s \leq 1$	-

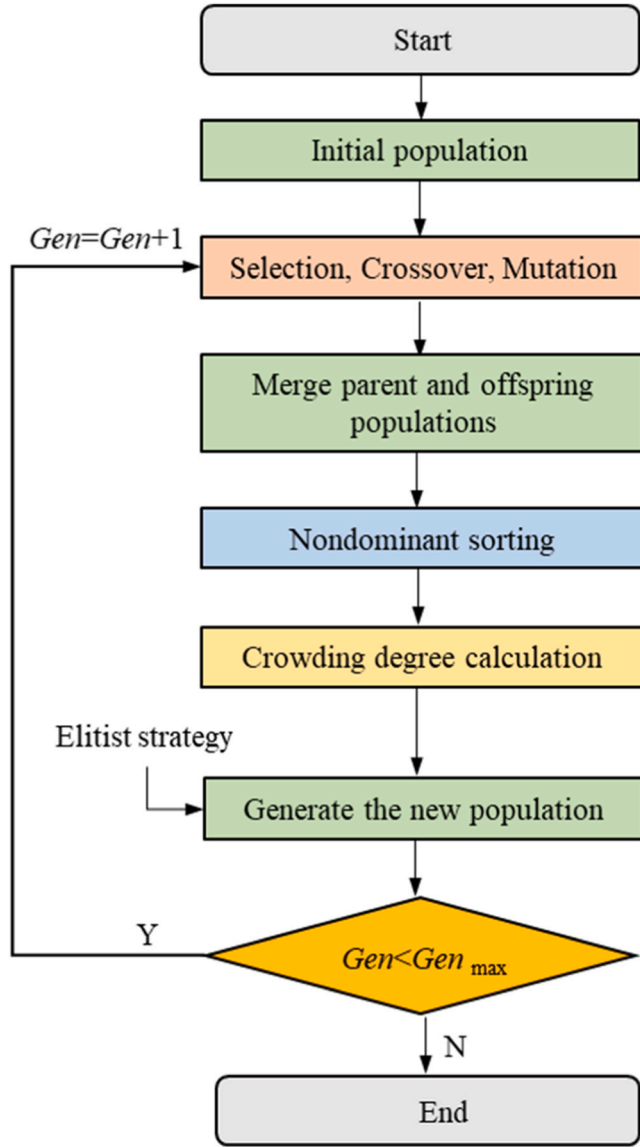


Fig. 10. The procedure of NSGA-II including integer variables.

6. Case study

6.1. Case background

To validate the feasibility and effectiveness of the proposed generative design framework for hybrid wind turbine towers, this section presents a comprehensive case study utilizing actual operational data from a large-scale wind farm located in eastern China. The investigation focuses on a representative 5 MW class wind turbine installation, with specific parameters for the wind farm and wind turbine carefully documented in Table 2. The concrete grade used for manufacturing the hybrid tower is C80, and the steel grade is Q355. For the prestressing system, high-tensile steel strands with a guaranteed ultimate tensile strength of 1860 MPa are implemented to provide the necessary compressive forces in the concrete sections. Notably, the interface between the nacelle and tower represents a critical design constraint. For this 5 MW turbine configuration, extensive industry experience and mechanical compatibility requirements dictate a standardized tower top diameter of 3.7 m.

Table 2
Wind farm and wind turbine parameters.

Type	Parameter	Value
Wind farm	IEC wind zone	III
	50 year maximum wind speed	32.5 m/s
	Annual average wind speed	6.4 m/s
	Wind shear coefficient	0.2
Wind turbine	Rated power	5 MW
	Nacelle mass	125 t
	Blade mass	110 t
	Nacelle moment of inertia	$1.58 \times 10^5 \text{ kg-m}^2$
	Blade moment of inertia	$9.99 \times 10^7 \text{ kg-m}^2$
	Blade diameter	193 m
	Number of blades	3
	Cut in wind speed	4 m/s
	Rated wind speed	11 m/s
	Cut out wind speed	25 m/s

6.2. Machine learning models

Using the Latin hypercube sampling method, 4096 hybrid towers were selected, and FE analysis was conducted to obtain structural responses for various design parameters. To eliminate the impact of varying ranges of feature parameters, normalization is applied to each feature. This process helps avoid significant differences in units, dimensions, and value ranges that could lead to a decline in machine learning performance. The normalization can be expressed using the following formula:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (21)$$

Where x represents the selected feature; $x_{\min}$ is the minimum value of the corresponding feature; and $x_{\max}$ is the maximum value of the corresponding feature.

This study employs the Pearson correlation coefficient to assess the independence of features and evaluate the quality of the dataset, as shown in Fig. 11. With the exception of the height of the concrete segment (H_c), the correlations among the remaining design variables fluctuate around zero, indicating that the input parameters are independent of each other. H_c inevitably exhibits a positive correlation with the total height of the tower (H), as the other design variables are generated within their minimum and maximum boundaries, while the H_c is constrained within H . Among all design variables, feature H exhibits the highest correlation with structural responses, followed by features t_{ct} and t_{sb} . The correlation between H and f_{1st} is the strongest, with a coefficient of -0.74 . Feature H also shows positive correlations with σ_{uct} , σ_{scb} , Δ_{max} , and θ_{max} , all of which are positively correlated. The feature t_{ct} exhibits the strongest positive correlation with σ_{ucc} , with a correlation coefficient of 0.35, while the feature H shows the strongest negative correlation, with a coefficient of -0.31 . Feature t_{sb} is most strongly correlated with the D_s , while the feature t_{ct} shows the strongest correlation with the D_c , both exhibiting negative correlations with coefficients of -0.25 and -0.31 , respectively. Additionally, feature t_{st} is most strongly correlated with σ_{sos} , while feature t_{sb} shows the strongest correlation with σ_{us} and σ_{scr} , with correlation coefficients exceeding 0.5.

Regression algorithms for three machine learning models were developed to address the first-order frequency, maximum tensile stress in the concrete section, maximum displacement and angle at the tower top, maximum stress in the steel section, maximum compressive stress in the concrete section, as well as local stability stress and overall stability check value of the steel section. Additionally, classification algorithms were developed to assess whether fatigue failure occurred of steel and concrete. The Mean Absolute Error (MAE) was chosen as the hyper-parameters optimization target for regression algorithms, while Accuracy was selected for classification algorithms. Table 3 lists the global optimal hyper-parameters for each model.

Table 4 compares the regression algorithms of three machine

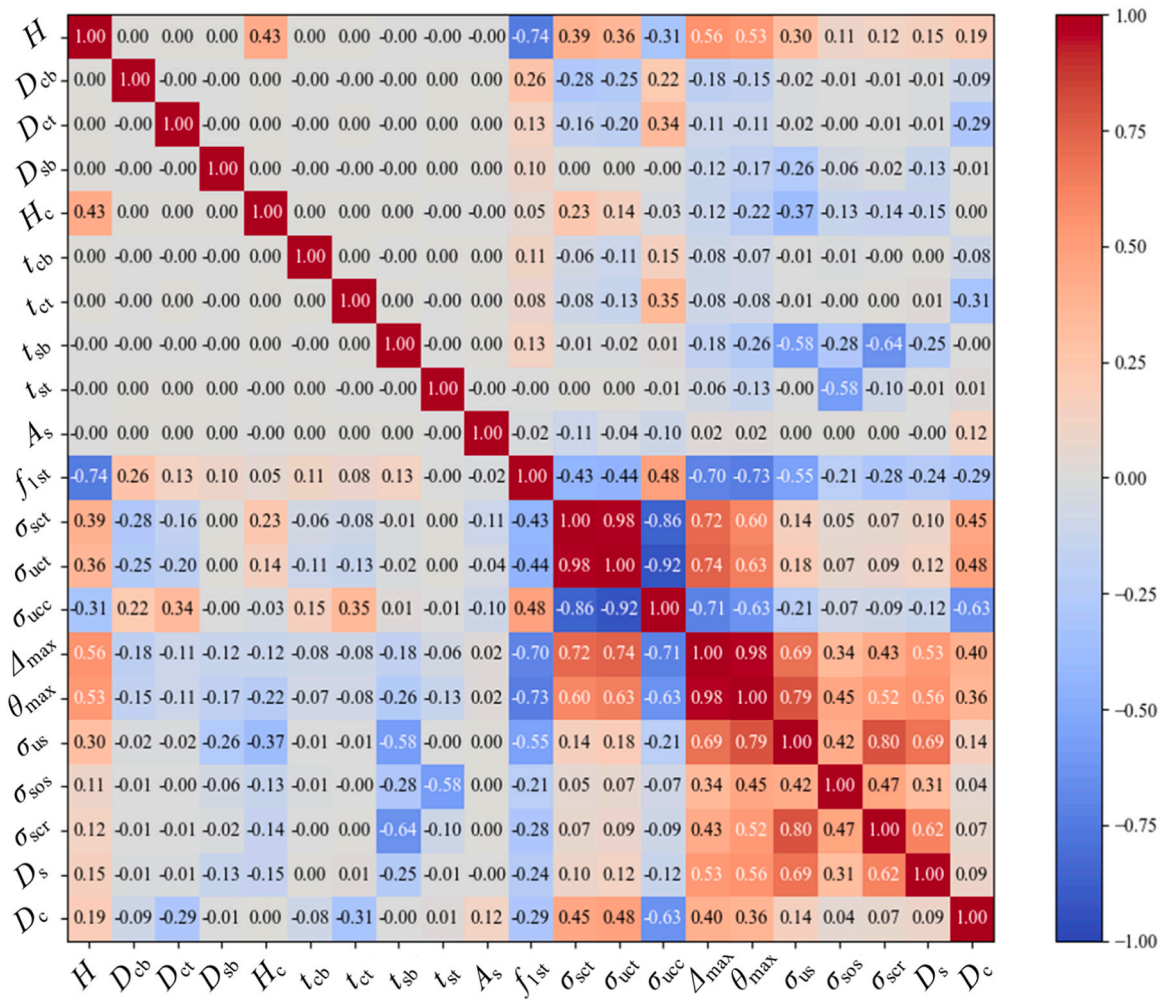


Fig. 11. Correlation coefficients between parameters.

Table 3
Model parameters of machine learning models.

Algorithms	Hyper-parameters	Regression algorithm parameter values	Classification algorithm parameter values
SVM	Punishment coefficient	1000	100000
	Kernel function parameter	0.1	0.001
ANN	Hidden Layers	25	150
	Initial learning rate	0.05	0.01
	Maximum number of iterations	200	200
XGBoost	Learning rate	0.1	0.2
	Tree by tree	100	100
	Maximum depth of a tree	5	5
	Minimum loss function descent value for node splitting	0.8	1.0

learning models based on R^2 , $RMSE$, and MAE . The results indicate that the ANN algorithm outperforms the other two, with a higher R^2 and significantly lower $RMSE$ and MAE , demonstrating the best fit for the samples. Although the XGBoost algorithm performed slightly less well in fitting concrete stress, it maintained an R^2 above 0.9 for other structural responses. In contrast, the SVM algorithm yielded the poorest fit for the tower's structural responses, particularly in predicting frequency and local stability load capacity of the steel. To provide a clearer perspective

of the prediction effectiveness of ANN model, Fig. 12 shows a comparison of the predicted values from the ANN regression algorithm against the measured values. It is evident that the ANN's structural response predictions are more concentrated within $\pm 10\%$ error margin.

Fig. 13 presents the confusion matrices for the machine learning classification algorithms on the test set. The diagonal numbers in the upper left four boxes represent the count of correctly classified samples, while the off-diagonal numbers indicate the number of misclassifications. The values in the bottom right box reflect the classification accuracy, which shows the proportion of correctly identified samples for both categories relative to the total samples. The first two boxes in the last row represent precision, indicating the proportion of correct predictions within each category, while the first two boxes in the last column reflect recall, representing the proportion of correctly predicted samples within the true samples. Higher recall and precision values indicate better classification performance, resulting in fewer misclassifications. Overall, all three models demonstrate strong classification effectiveness for fatigue failure, allowing for the selection of any model for classification tasks.

6.3. Optimization results

Among the three machine learning models, the ANN exhibited the best performance in predicting the structural responses of the hybrid tower. Consequently, the ANN model was used as a surrogate model for FE model to evaluate the automatically generated solutions. The key parameter settings for the NSGA-II algorithm were as follows: maximum

Table 4
Performance evaluation of structural response regression algorithms.

	SVM			ANN			XGBoost		
	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE
f_{1st}	0.7891	0.0517	0.0588	0.9772	0.0145	0.0187	0.9338	0.0095	0.0124
σ_{set}	0.8675	0.3346	0.5619	0.9904	0.2496	0.3896	0.9279	0.5082	0.8371
σ_{uct}	0.8703	0.7086	1.1345	0.9911	0.4336	0.6246	0.9333	1.1207	1.699
σ_{ucc}	0.8754	1.3296	2.0226	0.9901	0.8998	1.323	0.9425	2.1702	3.096
Δ_{max}	0.9850	51.957	73.7493	0.9920	46.1905	69.9422	0.9517	94.4908	132.304
θ_{max}	0.9662	0.0667	0.0824	0.9877	0.0315	0.0426	0.9473	0.0013	0.0018
σ_{us}	0.8980	3.5101	5.9349	0.9984	4.7529	6.9741	0.9868	10.2829	15.1414
σ_{sos}	0.9458	0.0458	0.0559	0.9787	0.0279	0.0368	0.9884	0.0129	0.0259
σ_{scr}	0.8248	0.1418	0.2314	0.9553	0.0938	0.1773	0.9795	0.0474	0.1207

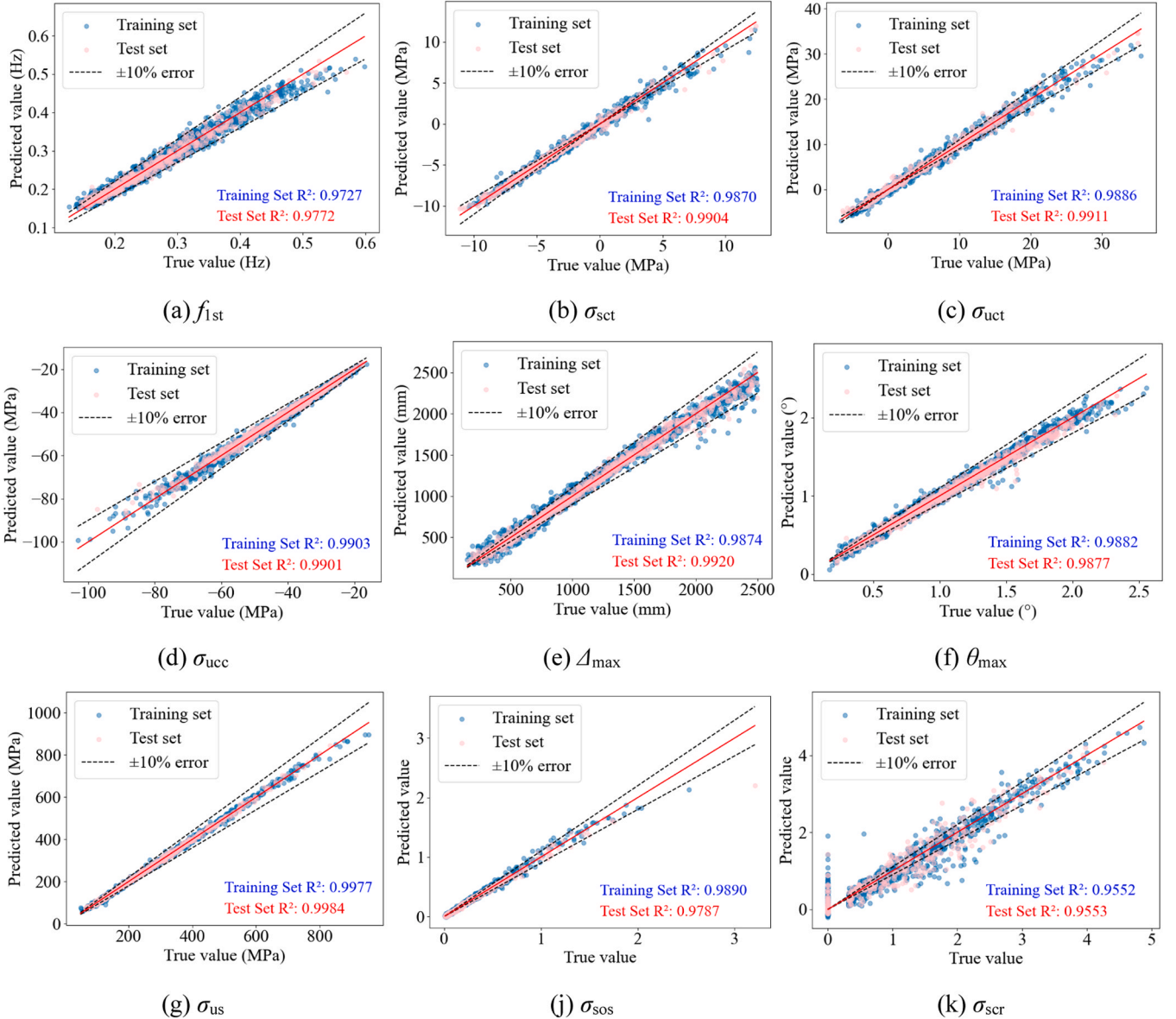


Fig. 12. Prediction results of ANN regression algorithm.

iteration count of 100, population size of 500, crossover rate of 0.9, and mutation rate of 0.01.

The NSGA-II algorithm provided the Pareto front, representing the optimal balance between AEP and construction costs. To validate the effectiveness of this generative design method, the Pareto front obtained from the surrogate-model-based generative design method was

compared with that obtained from traditional FE-model-based design method under the same design constraints, as shown in Fig. 14. Fig. 14 indicates that the Pareto fronts of generative design and traditional design are relatively close, particularly when employing ANN algorithms in generative design. This suggests that using appropriate machine learning models as an alternative for design evaluation is feasible.

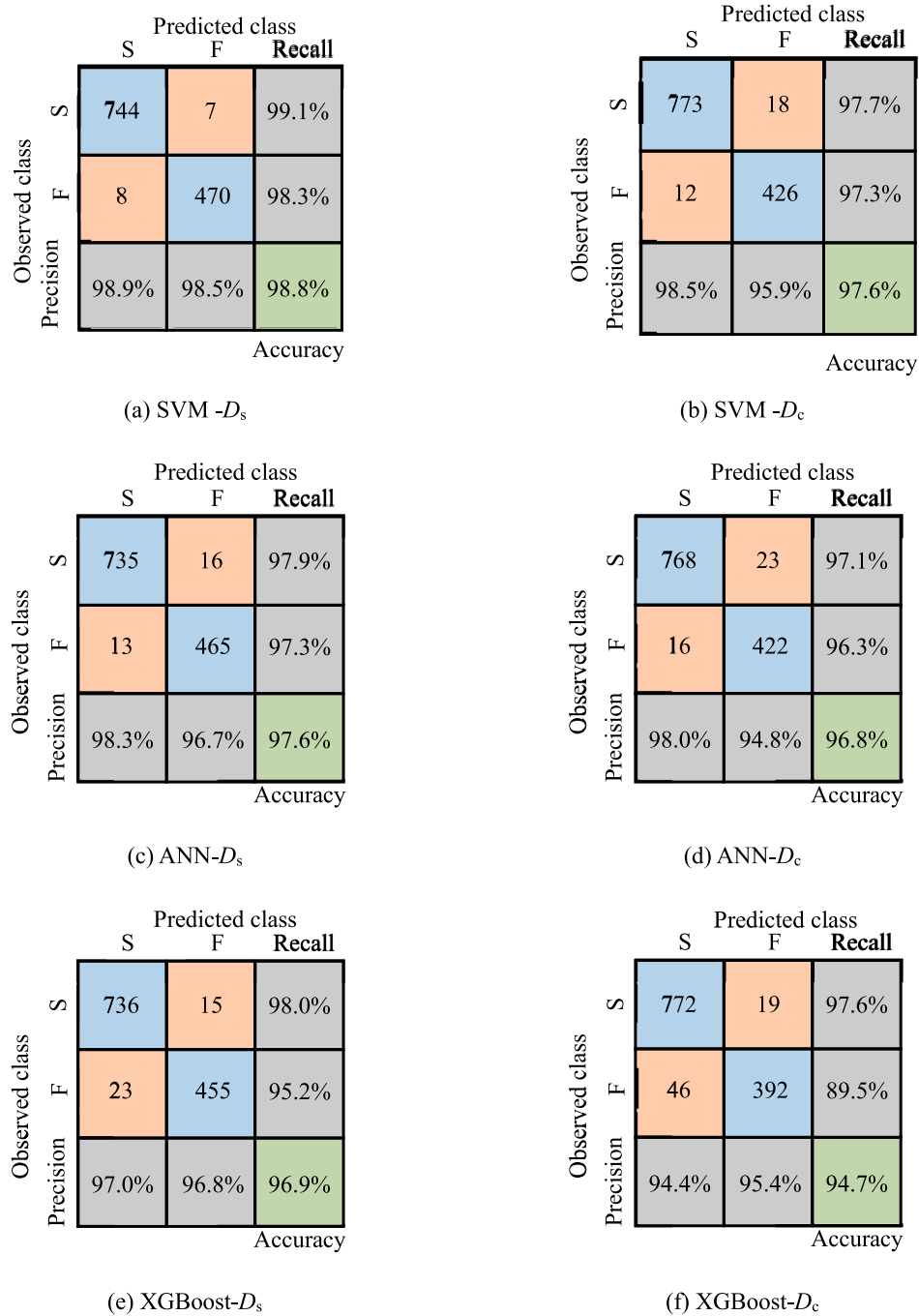


Fig. 13. Confusion matrix of classification results of fatigue damage. Notes: "S" indicates no fatigue failure, while "F" signifies the occurrence of fatigue failure.

A set of economically efficient design solutions was identified on the Pareto front for engineers' reference. These solutions are non-dominating, meaning no single solution outperforms another in both objective functions. The utopian point method [41] was employed to determine the best optimal solution to assist engineers in their decision-making process. Table 5 presents the optimal design results for both generative design and traditional design, including design variables, optimization objectives, and computation times. Notably, the generative design based on surrogate models significantly reduces optimization time, achieving an approximately 90 % increase in design efficiency. The tower parameters derived from the generative design were substituted into the FE model to recalculate structural responses, which were then compared with predictive results from the machine learning regression models, as shown in Fig. 15. All three machine

learning classification models indicate that the optimized tower structure will not experience fatigue failure, which is consistent with the FE analysis results. As illustrated, the SVM model showed unsatisfactory predictions for some structural responses, resulting in certain structural responses of the final optimized solution exceeding structural design limits. In contrast, the predictions from the ANN and integrated XGBoost models were relatively accurate, with structural responses of the final optimized solution closely approaching design limits, indicating effective material utilization in the optimized structure. The generative design based on the ANN model provided an optimization solution comparable to that of FE design. Additionally, the generative design based on the XGBoost model also yielded a commendable optimization solution, reducing construction costs by approximately 4.4 % while incurring a 1.13 % loss in AEP. Although the optimization time based on

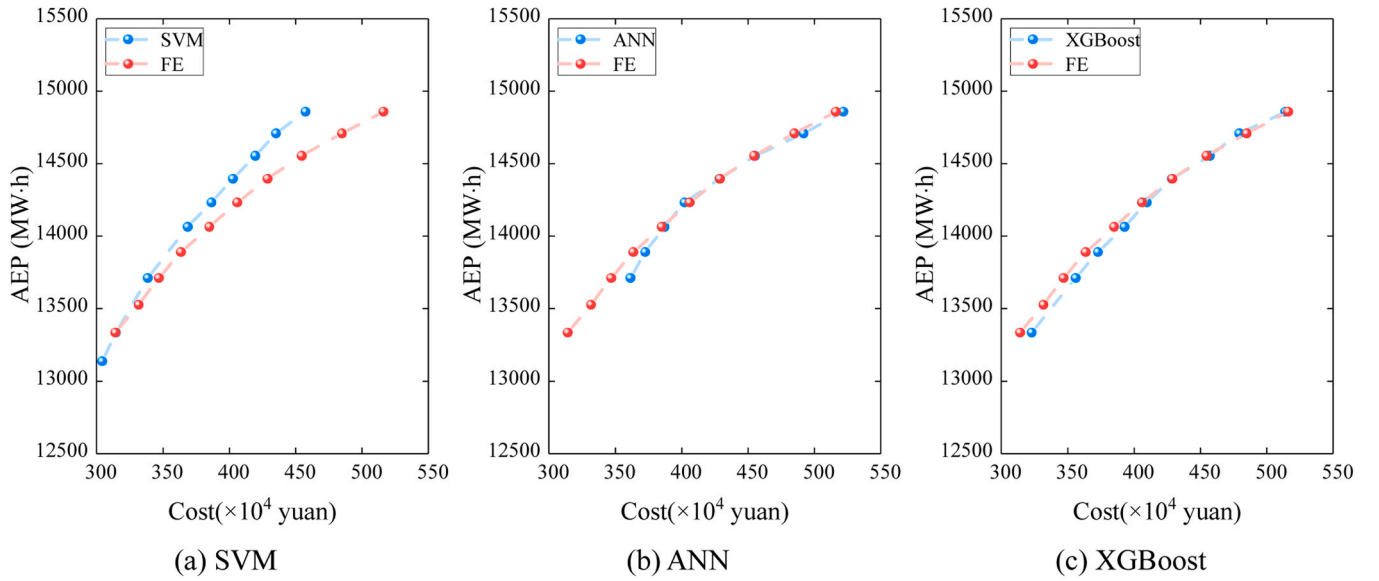


Fig. 14. Comparison of Pareto front of different algorithms.

Table 5
Optimization results of different algorithms.

Results	Symbols	Unit	FEM	SVM	ANN	XGBoost
Variables	H	mm	165000	165000	165000	160000
	H_c	mm	89100	103950	90750	83200
	D_{cb}	mm	9964	8946	9808	9590
	D_{ct}	mm	5860	5110	5630	5320
	D_{sb}	mm	4742	3710	4178	4270
	t_{cb}	mm	259	191	236	198
	t_{ct}	mm	210	302	230	284
	t_{sb}	mm	21	26	26	26
	t_{st}	mm	12	10	10	10
	A_s	mm ²	5187	4001	5393	5389
	Cost	$\times 10^4$ yuan	428.78	402.73	428.58	409.73
Objectives	AEP	MW·h	14395.96	14395.96	14395.96	14232.62
Number of structure analysis	N	/	50000	4096	4096	4096
Computing time	T	s	117340	7990	7898	7832

the traditional FE in this case is still within an acceptable range, in more complex optimization problems, the number of optimization variables and the scale of the design space may far exceed the scope of the current study, leading to a significant increase in optimization time. Therefore, the proposed Machine Learning (ML) method is not only suitable for the current problem but also provides a more scalable framework for more complex optimization scenarios.

7. Conclusions and future work

This study presents a generative design framework for steel-concrete hybrid towers based on machine learning and multi-objective optimization algorithms. The framework includes three modules: database construction, machine learning model development, and multi-objective optimization. Its aim is to automatically obtain the optimal design solution under specified design constraints. The following points can be drawn.

- (1) A FE model was developed to obtain the frequency, stress, and deformation of the SPCHT. A fatigue damage calculation framework based on wind speed probability distributions was developed to calculate the cumulative fatigue damage of materials.
- (2) Regression algorithms of three different machine learning models were used to predict the frequency, stress, and deformation of SPCHT, while classification algorithms were employed to

determine whether it experienced fatigue failure. The performance of three models was compared, revealing that the ANN-based model achieved the highest prediction accuracy, whereas the SVR-based model demonstrated the lowest accuracy.

- (3) The optimization variables included the total height of the tower, and a multi-objective optimization model for SPCHT was established with the objectives of maximizing annual energy production and minimizing construction costs. The NSGA-II optimization algorithm was used to achieve an optimal balance between construction costs and annual energy production.
- (4) Case study results indicate that the Pareto front obtained through generative design based on surrogate models is similar to that derived from traditional FE model-based design, but with approximately 90 % improvement in computational efficiency.

One limitation of the present study is that aerodynamic damping is not explicitly modeled within the structural dynamic framework. While the applied aerodynamic loads are obtained from Bladed simulations that inherently consider aeroelastic effects, including rotor-air interaction and associated damping, this indirect treatment may not fully capture the coupled dynamics in all scenarios. In future work, we plan to incorporate simplified aerodynamic damping models or establish a reduced-order aeroelastic coupling framework to better account for the dynamic interaction between the rotor and the supporting structure, especially under varying operating conditions.

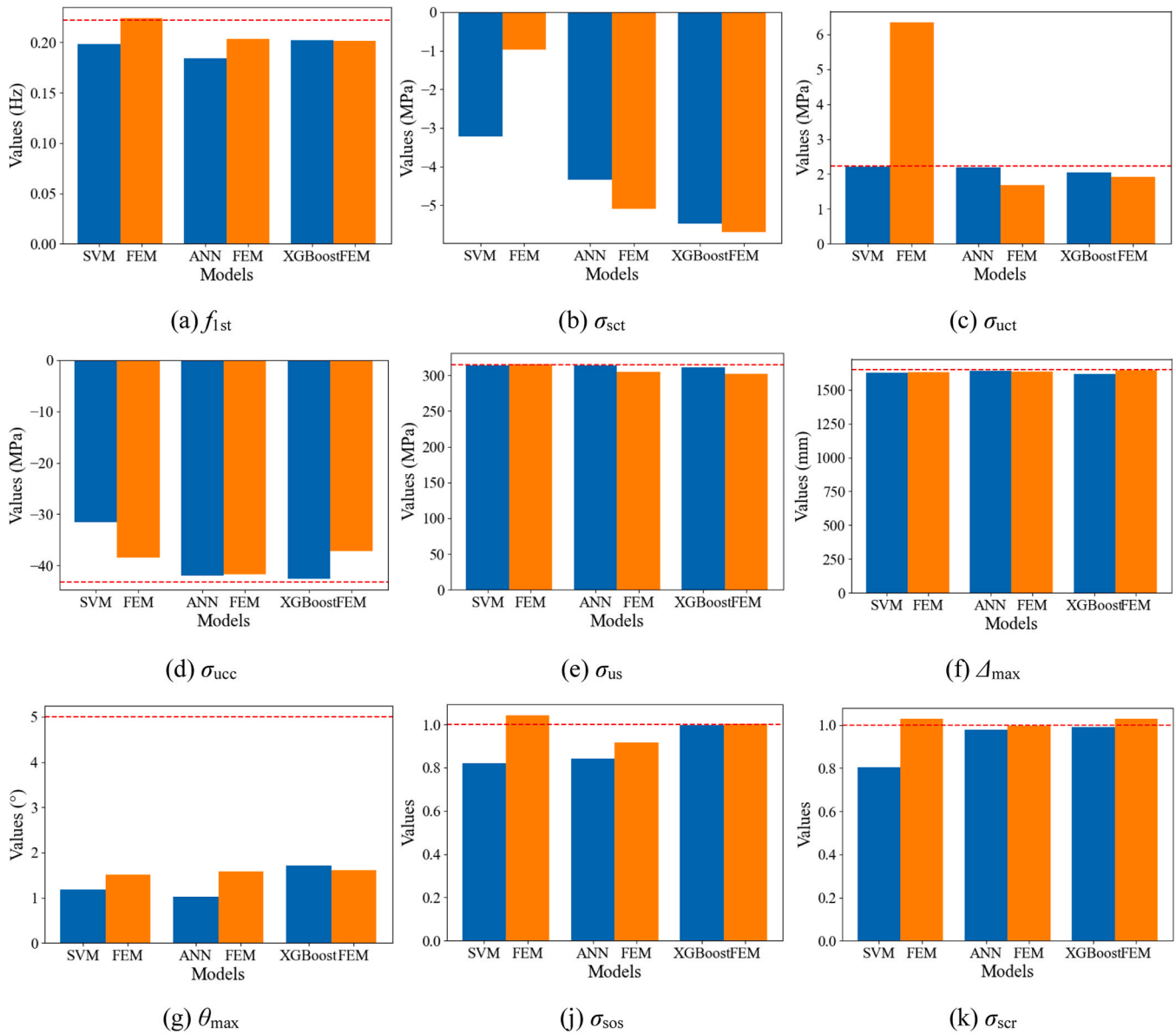


Fig. 15. Verification of structural responses of different algorithms.

CRedit authorship contribution statement

Yu Cheng: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Jiepeng Liu:** Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Hongtuo Qi:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Yannan Zhao:** Writing – original draft, Resources, Methodology. **Xuhong Zhou:** Validation, Supervision, Funding acquisition, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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